MIL-L-4352D
9 November 1973
Superseding
MIL-L-4352C(ASG)
20 Aug 1968

MILITARY SPECIFICATION

LENSES, GUN CAMERA, MOTION-PICTURE, 16MM FORMAT

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers 16 millimeter (mm) motion-picture gun camera lenses, with filters, for airborne application.

1.2 Classification. The lenses shall be of the following types and classes as specified (see 6.2).

Type	Nominal focal length (mm.)
I	12.5
II	17
III	25
IV	35
V	50
VI	75
VII	150

Class	
A	Gun sight aiming point (GSAP)
B	Threaded for "C" mount

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NNN-P-[illegible]0	Paper, Lens
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-665	Boxes, Paperboard, Metal Edged and Components
PPP-B-676	Boxes, Setup
PPP-C-843	Cushioning Material, Cellulosic

FSC 6760

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Military

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-F-9329	Filters, Light, Photographic, General Specification for
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-150	Photographic Lenses
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Ill., 60606).

American National Standards Institute

ANSI PH22.76	Threaded Lens Mounts for 16mm and 8mm Motion Picture Cameras

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction inspection (see 4.4).

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications and standards as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality and entirely suitable for the purpose intended.

3.4 Design. Lenses shall conform to type V of MIL-STD-150 and shall be as specified in 3.4.1 and 3.4.2.

3.4.1 Class A lenses shall be designed with gun sight aiming point (GSAP) mounting as shown on figure 1.

3.4.2 Class B lenses shall be designed with a threaded "C" mounting as shown on figure 2.

3.5 Optical characteristics. Lenses shall be designed to incorporate the optical characteristics specified in 3.5.1 through 3.5.9.

3.5.1 Focal length and aperture. Focal lengths and relative maximum apertures shall be in accordance with table I.

3.5.2 Format size. The lenses at all aperture settings shall focus a cone of light at the focal plane which will completely fill the standard 16 mm. motion picture format.

TABLE I. Focal lengths and apertures

Type	Nominal focal length	Equivalent focal length (EFL)	Relative maximum aperture	
			Class A	Class B
I	12.5 mm.	12.5 ±0.5 mm.	1/	1.4
II	17 mm.	17.0 ±0.8 mm.	1/	1.4
III	25 mm.	25.0 ±1.0 mm.	1/	1.4
IV	35 mm.	35.0 ±1.4 mm.	2.0	1.4
V	50 mm.	50.0 ±2.0 mm.	2.8	1.6
VI	75 mm.	75.0 ±3.0 mm.	1/	2.0
VII	150 mm.	150.0 ±6.0 mm.	1/	5.5

1/Not Available

* 3.5.3 Flange focal distance. The flange focal distance (FD) for class A lenses shall be 1.0185 ±0.0005 inches and for class B lenses 0.6900 ±0.0005 inch (see figures 1 and 2).

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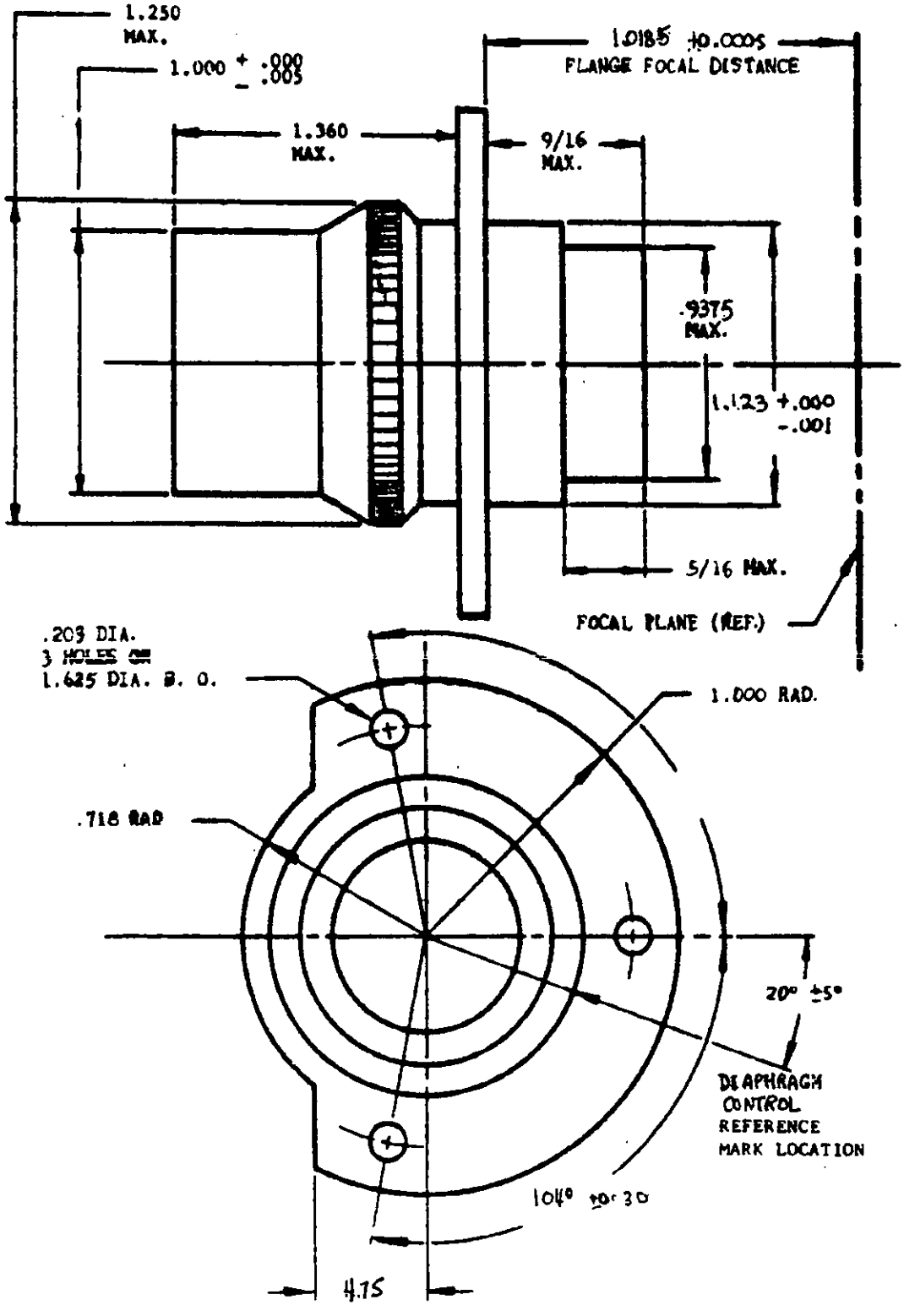


FIGURE 1. GSAP mount

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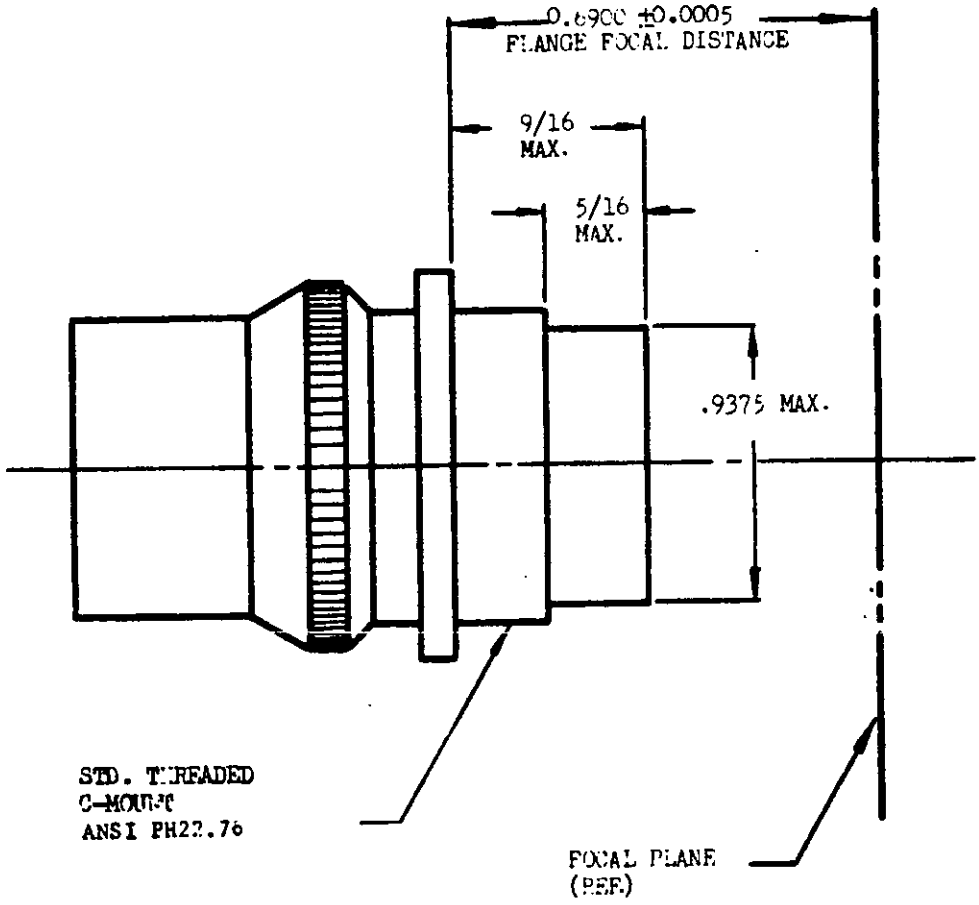


FIGURE 2. C-mount.

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* 3.5.4 Resolving power. The area weighted average resolution (AWAR), in line pairs per millimeter, in both radial and tangential directions, shall be tested in accordance with 4.6.2.3, and shall be as follows:

12.5 mm. nominal focal length	AWAR = 55
17 through 75 mm. nominal focal length	AWAR = 60
150 mm. nominal focal length	AWAR = 45

3.5.5 Color correction. The lens shall be color corrected for the "C" through "F" lines.

3.5.6 Distortion. When tested in accordance with 4.6.2.5, the maximum distortion of the lens shall not exceed 0.1 millimeter at any point in the specified field of view, except for 12.5 mm. and 17 mm. lenses which have a permissible dislocation not to exceed 0.2 millimeters at any point in the field of view.

3.5.7 Relative illumination. When tested in accordance with 4.6.2.7, the relative illumination shall be such that at f/2.8 the image illumination at angles "A" and "B" (formula Tan. A = 3.5/EFL and Tan. B = 6.5/EFL (EFL in millimeters) shall be not less than the following percentages. The relative illumination for EFL of 150 mm. shall be at f/4.5.

EFL	Percent at A	Percent at B
12.5 mm.	60	20
17 mm.	60	20
25 mm.	80	45
35 mm.	85	65
50 mm.	90	80
75 mm.	90	80
150 mm.	95	85

3.5.8 Transmittance. When tested in accordance with 4.6.2.8 the transmittance shall be at least 80 percent for all focal lengths.

3.5.9 Reflection-reducing coating. All glass air surfaces of the lens shall be coated with a reflection-reducing coating conforming to MIL-C-675.

3.6 Diaphragm. Unless otherwise specified, each lens shall be equipped with a suitable iris diaphragm calibrated and marked in the f-stop system specified in MIL-STD-150. The diaphragm adjusting ring shall have click stops at each setting.

3.6.1 Aperture marking. The lens shall be provided with a range of aperture markings in accordance with MIL-STD-150.

3.7 Filters. Each lens shall be furnished complete with a suitable filter mount and a yellow filter conforming to type III of MIL-F-9329.

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3.8 Lens caps. Each lens shall be fitted with two tight-fitting lens caps, which will not come off when subject to vibration in accordance with requirements of 3.11.5.

3.9 Finish. Exposed metal parts and mounts of the lens shall have a durable, firmly-bonded, dull-black finish.

3.10 Erectors. Image erectors, when required, shall be as specified by the procuring activity in accordance with requirements arising from special mounting problems (see 6.2).

3.11 Performance. The lens shall be capable of withstanding each of the following environmental conditions, with no impairment of operation or optical quality after being subjected to any one of combination of such conditions.

3.11.1 Temperature. When tested in accordance with 4.6.3.1 and 4.6.3.3, the lens shall be capable of withstanding temperatures in the range of -54° to +71° Centigrade (C) (-65° to + 160° Fahrenheit (F)).

3.11.2 Temperature shock. When tested in accordance with 4.6.3.4, the lens shall be capable of withstanding temperature changes, within a 5 minute interval, of -54° to +71°C (-65° to +160°F).

3.11.3 Altitude. When tested in accordance with 4.6.3.2, the lens shall be capable of operating at barometric pressure ranging from 30 inches of mercury (Hg) down to 3.4 inches Hg.

3.11.4 Acceleration. When tested as specified in 4.6.3.9, the lens shall be capable of withstanding and recovering from accelerations of 8g.

3.11.5 Vibration. When tested as specified in 4.6.3.10, the lens shall be capable of withstanding and recovering from a vibration of 15g about each of the three axes.

3.11.6 Humidity. When tested as specified in 4.6.3.5, the lens shall be capable of withstanding ambient temperatures of -18° to + 50°C (0° to +120°F) in combination with humidities of 10 to 90 percent.

3.11.7 Dust. The lens shall be capable of withstanding the dust test specified in 4.6.3.8.

3.11.8 Fungus. When tested as specified in 4.6.3.6, the lens shall be capable of resisting conditions conductive to fungus growth when tested in accordance with Method 508 of MIL-STD-810.

*3.11.9 Salt fog. The finish on the lens shall be capable of resisting the salt fog test specified in 4.6.3.7.

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3.12 Weight. The weight of the lens shall be kept to a minimum, consistent with the other requirements specified herein.

3.13 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be in accordance with MIL-D-1000.

3.14 Identification of product. The lens shall be permanently and legibly marked for identification with the following information in addition to that specified in 3.6.1:

(a) _____mm. (applicable focal length)
(b) Manufacturer's name
(c) Manufacturer's part number

3.15 Workmanship. The lens shall be constructed in a thoroughly workmanlike manner. Minor beauty defects, defined in MIL-STD-150, shall not be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.3 Test conditions. Unless otherwise specified, examination and tests shall be performed in standard atmospheric conditions (ambient room temperature, relative humidity, and barometric pressure).

4.4 Preproduction inspection. Preproduction inspection shall consist of the examination and tests specified in 4.6.

4.4.1 Test sample. The test sample shall consist of one lens representative of the production lenses.

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4.4.2 Test reports. When required by the procuring activity (see 6.2), the contractor shall furnish a reproducible and three complete copies of the preproduction test reports in accordance with MIL-STD-831.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the following:

(a)	Individual tests	(see 4.5.1)
(b)	Sampling tests	(see 4.5.2)

4.5.1 Individual tests. Each lens shall be subjected to the examination specified in 4.6.1.

4.5.2 Sampling tests. One lens, randomly selected from each 10 production lenses (or fractional part thereof), shall be subjected to the performance tests specified in 4.6.2.

4.5.2.1 Rejection and retest. When one or more items from a lot fail to meet this specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the procuring activities the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.6 Inspection methods.

4.6.1 Examination of product. Each lens shall be visually examined to determine compliance with requirements of this specification in respect to workmanship, identification and operational markings, reflection-reducing coatings, finish, and for operation of iris diaphragm without binding.

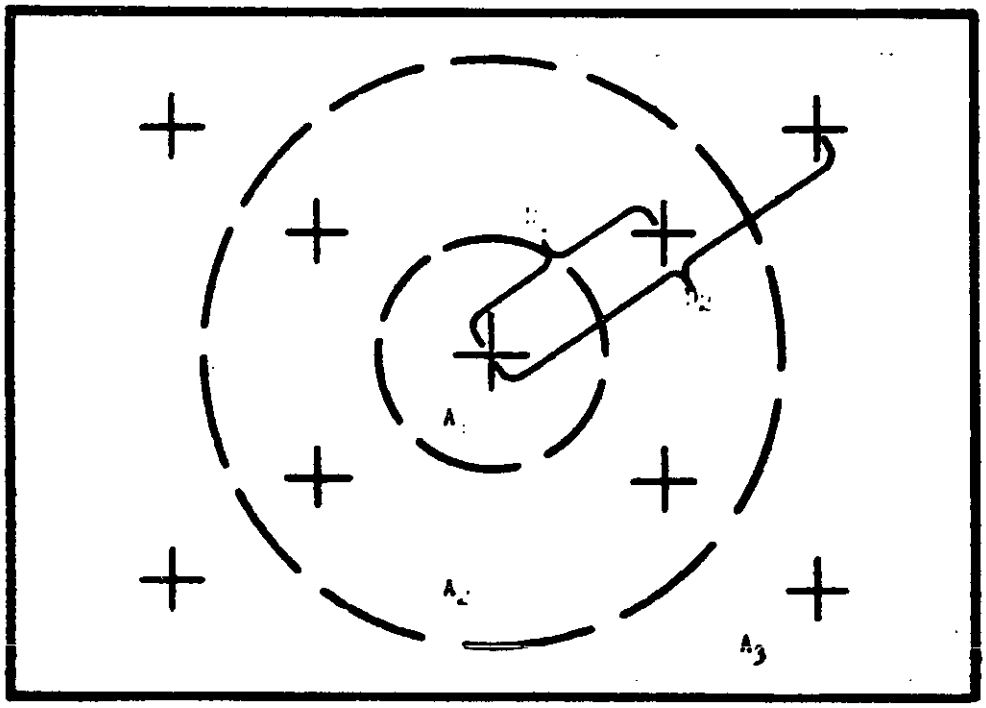
4.6.2 Performance tests. Each lens, selected at random as specified in 4.5.2, shall be subjected to the following tests.

4.6.2.1 Photographic. A photographic test shall be made with each selected lens. The test apparatus shall conform to that described in MIL-STD-150. A flash discharge tube shall be used to illuminate the target. A high-contrast target, with light lines on a dark background, shall be used. Film having an ASA speed of 50 shall be used, and it shall be developed to a gamma of 0.60 $\pm$0.10. This test shall be the basic test from which the EFL and FD are determined.

4.6.2.2 Flange focal distance. The flange focal distance test shall be performed as outlined in MIL-STD-150.

* 4.6.2.3 Resolving power. The value for resolving power (AWAR) shall be determined by exposing a 10.41 mm. by 7.47 mm. format with nine resolution target patterns located as shown on figure 3. The test shall be performed in accordance with Method 11 of MIL-STD-150, except that the reference to 1 1/4 degree increments shall be disregarded and the positions as shown in figure 3 be substituted. All exposures shall be taken at the same focus. The value for each tangential and radial resolution value shall be determined by averaging the four readings for each in the same zone or area.

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A_1 - [illegible] MM^2

A_2 - 33.56 MM^2

A_3 - 49.3[illegible] MM^2

D_1 = [illegible] MM (TYP. 4 PLACES)

D_2 = [illegible] MM (TYP. 4 PLACES)

+ REFERS TO A USAF 1951 RESOLUTION TARGET LOCATION

Figure 3. Target patterns

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4.6.2.4 Equivalent focal length. The equivalent focal length (EFL) shall be measured by means of Method 1 of MIL-STD-150, and the plane of highest AWAR shall be the criterion for its determination.

4.6.2.5 Distortion. Distortion shall be measured by means of one of the following methods in accordance with MIL-STD-150:

(a) Method 25
(b) Method 26
(c) Method 27
(d) Method 28

4.6.2.6 Relative aperture. The relative aperture shall be computed by the ratio of the measured EFL to the diameter of the effective aperture. Effective aperture shall be measured by Method 3 or Method 4 of MIL-STD-150.

4.6.2.7 Relative illumination. The relative illumination shall be measured by Method 7, Method 8, or Method 9 of MIL-STD-150.

4.6.2.8 Transmittance. The transmittance shall be measured by Method 5 or Method 6 of MIL-STD-150.

4.6.3 Preproduction tests. A sample lens, selected as specified in 4.4.1, shall be subjected to the environmental tests specified in 4.6.3.1 through 4.6.3.10. Reference to the lens shall be understood as including the filter and barrel with adjustable iris diaphragm. No tests of optical performance are required. Only the mechanical operation of the iris diaphragm shall be checked.

4.6.3.1 High temperature. The lens shall be subjected to the high temperature test, Method 501, Procedure I of MIL-STD-810, except that the period of exposure shall be reduced to 24 hours.

4.6.3.2 Altitude. The altitude test shall be performed in accordance with Method 500, Procedure II of MIL-STD-810. The minimum pressure shall be 3.4 inches of mercury.

4.6.3.3 Low temperature. The lens shall be subject to the low temperature test, Method 502 of MIL-STD-810. Minimum temperature during this test shall be -54°C (-65°F).

4.6.3.4 Temperature shock. The lens shall be subjected to temperature shock, Method 503, Procedure I of MIL-STD-810.

4.6.3.5 Humidity. The lens shall be subjected to the humidity test, Method 507, Procedure I of MIL-STD-810.

4.6.3.6 Fungus resistance. Fungus-resistance tests in accordance with Method 508 of MIL-STD-810 shall be performed only if the lens assembly contains parts of materials which are nutrients for fungi. The lens caps shall be in place on the lens when fungus-resistance tests are performed.

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4.6.3.7 Salt fog. Salt fog tests in accordance with Method 509 of MIL-STD-810 shall be performed only when required by the procuring activity (see 6.2).

4.6.3.8 Dust. The lens shall be subjected to the dust test, Method 510 of MIL-STD-810. Lens caps shall be in place on the lens during the test. Upon completion of the test, external glass surfaces shall be inspected for damage in addition to inspection of mechanical operation of the iris diaphragm.

4.6.3.9 Acceleration. The lens shall be subjected to the acceleration test, Method 513, Procedure I of MIL-STD-810. Acceleration shall be 8g, measured at the center of gravity of the lens.

4.6.3.10 Vibration. The lens shall be subjected to the vibration test, Method 514, Procedure I of MIL-STD-810.

4.7 Preservation, packaging, and packing. Preparation for delivery shall be inspected for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each lens assembly shall be preserved in accordance with Method IC-3 or IC-4 of MIL-P-116, with the following exceptions:

(a) Contact preservative shall not be used.
(b) Each lens assembly, after cleaning, shall be wrapped in four or more layers of lens tissue conforming to NNN-P-40.
(c) Cushioning shall be a minimum of 1-1/2 inches thickness. Material shall conform to PPP-C-843.
(d) Unit quantity for packaging shall be one lens assembly.
(e) Packaging for Method IC-3 preservation shall be each lens assembly, wrapped, cushioned, sealed, and placed in a snug-fitting box or carton conforming to PPP-B-676 or PPP-B-665.
(f) Packaging for Method IC-4 preservation shall be as specified in MIL-P-116.

5.1.2 Level C. Each lens assembly shall be packaged to afford adequate protection from physical damage during shipment to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Lens assemblies preserved and packaged as specified in 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-601 or PPP-B-621. As far as practicable, exterior containers shall be of uniform shape and size, minimum cube and tare consistent with the protection

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required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 100 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto.

5.2.2 Level C. Lens assemblies, preserved and packaged in accordance with 5.1.2 shall be packed in exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

LENSES, GUN CAMERA, MOTION-PICTURE, 16MM FORMAT
Specification MIL-L-4352D
Type and class_____(as applicable)
Focal length_____(as applicable)
Designation_____(as applicable)

6. NOTES

6.1 Intended use. These lenses are intended for use in 16mm. motion-picture gun cameras.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class (see 1.2).
(c) Data on erection system, if required (see 3.10).
(d) If preproduction test reports are required (sec 4.4.2).
(e) If salt fog test is required (see 4.6.3.7).
(f) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 The margin of this specification is marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Army - EL
Air Force - 70

User:
Army - AV

Preparing Activity:
Air Force - 70

Project No. 6760-0495

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

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